AGENTBIO — SAVED HYPOTHESIS REPORT

The MW effect and the XLogP effect interact: a size-lipophilicity mismatch (large AND lipophilic) produces impedance mismatch and disproportionately poor repurposing.

AUDIT NUMBERS — FROM THE REGISTRY AT SAVE TIME

FRAME	TEST	OR	95% CI	N	DISCOVERY RAW P	FDR Q	CONFIRM RAW P
narrow	logistic_interaction	0.999	[0.997, 1]	2186	4.52e-1	6.85e-1	–
broad	logistic_interaction	0.997	[0.996, 0.998]	3017	4.85e-9	3.84e-8	4.06e-16

NARRATIVE — CLAUDE OPUS 4.8

Statistical Finding Report: MW × XLogP Interaction on Repurposing Success

Hypothesis ID: run-09c577de-H08

Hypothesis

The hypothesis states, verbatim: "The MW effect and the XLogP effect interact: a size-lipophilicity mismatch (large AND lipophilic) produces impedance mismatch and disproportionately poor repurposing." This hypothesis was proposed by **Opus**, drawing on the source domain of **impedance matching in transmission-line power transfer**. The stated mechanistic justification is that "Simultaneous high mass and high lipophilicity maximizes reflected 'power' (sequestration + poor solubility), the opposite of a matched load."

Prior-literature tag (web-search, informational): UNCLEAR.

Bisociative provenance

Run ID: `run-09c577de`. Both generators proposed multiple source domains in this run.

Opus proposed:

- impedance matching in transmission-line power transfer
- loss aversion / endowment effect in behavioral decision theory
- quorum sensing thresholds in bacterial gene regulation
- stigmergy / pheromone-trail reinforcement in ant foraging
- vestibular sensory reweighting under conflicting cues (multisensory integration)

Sol proposed:

- cycle-life dominance in rechargeable batteries under repeated duty cycles
- impedance matching by transformer networks in radio-frequency circuits
- kinetic proofreading under near-cognate T-cell receptor recognition
- mode matching at refractive-index discontinuities in optical waveguides

Lead reviewer consolidation (from lead_decisions): The lead_decisions object records the two entries corresponding to this hypothesis (both under the impedance-matching-in-transmission-line domain, Opus). Neither entry contains an explicit READY / NEEDS_ENRICHMENT / DISCARDED status token; instead each records the effect estimate and restates the mechanism. The two recorded decisions are:

- OR=0.999, CI[0.997,1], n=2186 — rationale: "Simultaneous high mass and high lipophilicity maximizes reflected 'power' (sequestration + poor solubility), the opposite of a matched load."
- OR=0.997, CI[0.996,0.998], n=3017 — rationale: same as above.

The explicit READY/NEEDS_ENRICHMENT/DISCARDED label is not present in FACTS for this hypothesis's lead_decision entries, so it cannot be reported. (For context, the lead_decisions list also contains SKIPPED, SALVAGEABLE, and "not tested" designations for other hypotheses in this run.)

How this was actually tested

The literal computable proxy applied to the dataset, verbatim from FACTS.how_tested:

"interaction: whether the effect of [molecular weight (pubchem_mw), used as a continuous predictor] on repurposing success DIFFERS across the two groups defined by [XLogP lipophilicity (pubchem_xlogp) >= 5 (binary split)] (logistic base:moderator term)"

This mechanical interaction term — not the broader "impedance mismatch" biological/physical claim — is what the statistics below actually measured.

Statistical evidence underneath the p-value

FRAMING	TEST TYPE	ODDS RATIO	95% CI	N	DISCOVERY RAW P	FDR Q
narrow	logistic_interaction	0.999	[0.997, 1.0]	2186	0.45202263650678876	0.6846813464735183
broad	logistic_interaction	0.997	[0.996, 0.998]	3017	4.850091148208065e-09	3.842764525118698e-08

- **narrow:** The interaction-term odds ratio of 0.999 is essentially at 1.0, with a CI [0.997, 1.0] that reaches 1.0, indicating a negligible and non-significant interaction effect for this framing.
- **broad:** The interaction-term odds ratio of 0.997 is just below 1.0 with a CI [0.996, 0.998] that excludes 1.0, indicating a statistically detectable but very small negative interaction (each unit change in the interaction term is associated with slightly lower odds of repurposing success).

Discovery vs. confirmation

FRAMING	DISCOVERY RAW P	FDR Q	DISCOVERY PASS	CONFIRMATION RAW P	CONFIRMATION PASS
narrow	0.45202263650678876	0.6846813464735183	false	null	false
broad	4.850091148208065e-09	3.842764525118698e-08	true	4.064550550631848e-16	true

Passing confirmation means the effect reproduced on holdout data never used during discovery. Only the **broad** framing passed both discovery (FDR-corrected) and independent holdout confirmation. The **narrow** framing did not pass discovery, and its confirmation p-value is null (confirmation not applicable / not recorded), so it did not pass confirmation.

Confound checks

FACTS.confound_check is **null**. No confound-check object was persisted for this hypothesis. Therefore:

- No named alternative explanations are available to report.
- No unadjusted-versus-adjusted odds ratios, adjusted CIs, adjusted p-values, or `survives_adjustment` flags exist in FACTS for this finding.

Because the confound_check field is null rather than populated, it is not possible to state whether the effect survived adjustment for any confounder — no confound was recorded as tested, and none can be fabricated here. This is a material limitation of the auditable record for this hypothesis.

Verdict: The confound analysis cannot be evaluated. FACTS contains no computable confound results (confound_check = null), so no statement can be made about whether the effect survived adjustment for any alternative explanation. The absence of these checks should be treated as an open gap, not as evidence that the effect is robust to confounding.

Bottom line

The finding is a statistically detectable interaction, in the broad framing, between molecular weight (continuous) and an XLogP≥5 binary split on repurposing success in this curated dataset, with an interaction-term odds ratio of 0.997 (95% CI [0.996, 0.998], n=3017), which passed both FDR-corrected discovery (q = 3.84e-08) and independent holdout confirmation (p = 4.06e-16). This is a statistical association only — it is not evidence of a causal, biological, or clinical "impedance mismatch" mechanism, and the narrow framing showed no such effect (OR 0.999, non-significant). The strongest remaining caveat is that no confound checks were recorded for this hypothesis (confound_check = null), so the possibility that the association is explained by an unmodeled third variable has not been ruled out.